

Predict Once, Track Forever: The Trajectory-Predictability Privacy Gap in Direct-to-Cell

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Direct-to-cell (D2C) cellular service over low-earth-orbit satellites has entered commercial operation, beginning with T-Mobile in the United States, and its subscriber base is growing fast. Its base station is not a tower whose location an operator can keep private; it is a satellite whose ephemeris is broadcast on public feeds and embedded verbatim in SIB-19. A passive radio receiver outside the operator can predict, deterministically and hours to a day ahead with kilometer-scale accuracy, where every D2C cell will be. We measure what this buys an adversary. Five minutes of passive observation localize a UE to a median of 65 km across thirty cities; twenty minutes drop the median below 50 km. A timing-advance side channel tightens the five-minute median to a 1 to 2 km regime (median 1.8 km, P90 below 4 km), bounded jointly by timing-advance quantization and public-ephemeris accuracy. We then test the two operator-side privacy architectures the cellular community most often cites, PGPP and LOCA, and show neither reaches the threat: the attack input is disjoint from what either defense modifies. Three ad-hoc operator patches fail for the same structural reason. We close with CLOAK, a design sketch that adapts the V2X mix-zone primitive to orbital geometry through three composable mechanisms (cohort-synchronized handover, multi-satellite attach ambiguity, timing-advance quantization), and state the boundary conditions it would have to respect.

1 Introduction

Transmitting to a satellite has long carried a targeting risk: operators of satellite terminals in conflict zones have been warned that uplinking can expose their position to whoever is listening. Direct-to-cell changes the calculus. Revealing a user's location no longer requires breaking a link, spoofing a signal, or cooperating with the operator. The base station's trajectory is already public, and the user reveals itself simply by attaching to it.

Direct-to-cell (D2C) cellular over LEO is now in commercial deployment. SpaceX has launched the first generation of a D2C-capable Starlink constellation, and T-Mobile has offered commercial D2C service in the United States since 2025; the first independent measurement study confirms the service is live, observable, and expanding with the deployed satellite count [12]. The radio access follows the 3GPP Release 17 NTN extensions in TS 38.331 [3]: a UE attaches to a passing satellite, hands over every minute or so as the source descends below the link budget, and registers against a stateful core that follows it across orbital passes.

The leakage. A satellite's position is public. Anyone with a CelesTrak account [21] and an SGP4 propagator [41] can compute, to within about a kilometer and hours to a day ahead, where any D2C satellite will be at any past or future timestamp.¹ The same input is broadcast to UEs in SIB-19. The 3GPP standard treats this disclosure as a service primitive: UEs need it to plan acquisition. We treat it as the surface a passive eavesdropper exploits to infer where a UE is and where it will be.

The first work to operationalize this leakage is Liu et al.'s DCATOR analyzer [26] at IEEE S&P 2025, which localizes active D2C UEs by monitoring broadcast signaling and combining it with public trajectories. We take the threat as established. Three questions DCATOR raises but does not answer are this paper's spine. First, how does the adversary's fit error scale with observation window and latitude across a city-scale benchmark? Second, does the timing-advance value from the random-access response tighten the fit, and by how much? Third, do PGPP and LOCA, the two

¹Public two-line elements with SGP4 are accurate to roughly a kilometer near epoch and degrade by about 1 to 3 km per day [22]. An adversary that instead tracks the satellite's own downlink can refine this to sub-100 m (§2.2); we keep the public two-line-element accuracy as the conservative baseline.

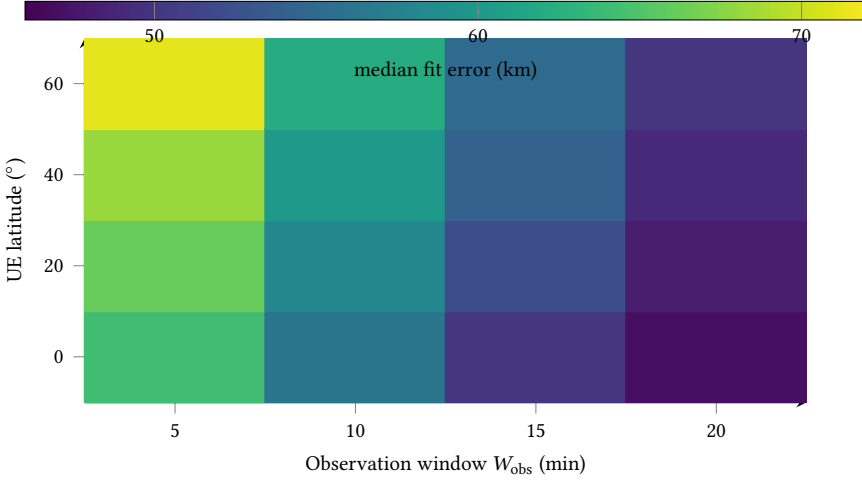


Fig. 1. Median fit error (km) over UE latitude and observation window.

operator-side architectures the cellular-privacy community most often cites, reach this adversary? Our answer to the third is no, and it is the load-bearing claim: the defenses the community reaches for protect the user’s identity, while the attack only consumes when and where the user attaches.

What we measure. Figure 1 answers the first question across thirty globally distributed cities and an 86-point latitude-longitude sweep. Five minutes of passive observation localize a UE to a 65 km median. Twenty minutes drop the median below 50 km, with the lowest error at mid-latitudes where serving-satellite handovers are most frequent. A second channel, the timing-advance value reported during random access, tightens the five-minute median to a 1 to 2 km regime (median 1.8 km, P90 below 4 km). We confirm that both inputs the attack consumes, the broadcast ephemeris and the timing advance, are recoverable from a real downlink, not only from our fit model (§3.1).

Why existing defenses do not reach. PGPP [36] replaces the UE’s permanent identifier with anonymous tokens issued through an MVNO, so the base station never sees a stable identifier. LOCA [29] splits the operator into a broker that knows identity and an infrastructure provider that knows location, so neither party learns both. Both target the operator. The D2C attack does not read identity, does not contact the operator, and does not depend on which entity inside the operator sees what. We deploy each architecture in our simulator and measure the adversary’s fit error against it. The curves sit exactly on the baseline (§4).

Why ad-hoc patches do not reach either. A natural reaction is to suppress the leak with operator knobs. We test four (§5). Noisy SIB-19 ephemeris breaks the UE before it touches the adversary, who reads Celestrak. Dummy attach traffic at footprint scale requires dummies to follow plausible UE trajectories, which are themselves localizable. TLE withholding does not work because the orbit is reconstructible from optical and signal-of-opportunity observation. Encrypted SIB-19 fails because anyone can buy a D2C phone. Each fails for a distinct, stated reason.

Toward something that reaches. The mature analog is the V2X mix-zone literature [5, 10], which has spent two decades on a closely related problem: globally observable mobility paired with a need for anonymity. §6 sketches CLOAK, three primitives interposed between the public trajectory and the UE cohort. CLOAK is a design direction, not an evaluated system; we state the cost and boundary condition of each primitive.

Contributions.

- **A predictive-horizon measurement.** We quantify adversary fit error across 86 (latitude, longitude) UE points and six observation windows, and we measure the timing-advance side channel, which tightens the five-minute median to a 1 to 2 km regime (§3).
- **A structural-invariance result.** We show that PGPP and LOCA leave the adversary’s fit unchanged, and we name the reason: input-disjointness, not defense deficiency. Three ad-hoc patches fail for the same reason (§4, §5; Table 1).
- **A real-signal grounding.** We validate the public trajectory input with a real Starlink catalog and SGP4, and decode SIB-19 ephemeris and a broadcast timing-advance value from a controlled Release-17 downlink, confirming the side channel is observable in practice (§3.1).
- **A scoped defense direction.** We sketch CLOAK with the cost and boundary condition of each primitive, and name across-cohort correlation as the open question any follow-up must answer (§6, §7).

2 Background and Threat Model

2.1 What a D2C UE does on the air

A D2C UE follows the 3GPP NR-NTN access procedure [1–3]. It reads synchronization and system information from the serving satellite, computes timing-advance compensation from the broadcast ephemeris, attaches via a long-RTT-aware random-access procedure, and is handed over to the next satellite roughly every minute as the source descends below the link-budget threshold. The handover decision uses RSRP measurements and the SIB-19 ephemeris [8, 27].

The property that matters for privacy is that SIB-19 is broadcast unprotected to every UE in cell. It carries the serving satellite’s orbital state, refreshed within hours of any maneuver, and is consistent with what CelesTrak publishes. A UE with no Internet connection still knows where the serving satellite will be at any time within the SIB-19 validity period. So does anyone with a software-defined radio.

2.2 Threat model

The adversary is a passive radio receiver placed anywhere on Earth, with three capabilities. It decodes broadcast D2C signaling (SIB-19, the master information block, the random-access response) within reception range. It has a public TLE feed and a standard SGP4 propagator. It has compute for offline correlation. We assume it does not break cryptographic primitives.

The adversary is not the operator and is not part of any MVNO, and it need not contact either. It need not share a continent with the UE: signal-of-opportunity studies have tracked the Starlink downlink with consumer hardware to sub-100 m accuracy [18, 33, 34], which means the satellite trajectory is independently observable even when the public catalog is withheld (§5). For the localization numbers we report, we use only the public catalog and SGP4, the weaker and more conservative input.

This adversary differs from the one PGPP and LOCA target. Both architectures treat the operator, or a colluding subset of it, as the party that must be denied the joint identity-location pair, and design backend protocols accordingly. The D2C adversary sits outside the operator and reads the broadcast channel. No backend rearrangement reaches it.

3 The Predictive Horizon

We measure how much passive observation suffices to localize a UE well enough to predict its future attach pattern.

Setup. We build a 4,408-satellite Walker-delta constellation matching SpaceX’s FCC-authorized Starlink Gen-1 license (SAT-MOD-20200417-00037 [38]): five shells at five distinct (inclination,

altitude) pairs. We propagate every satellite at 10-second resolution with SGP4's vectorized API [41]. For each of 30 globally distributed cities plus a 14-latitude by 4-longitude sweep (86 sample points), we compute the truth, a serving-satellite timeline at no less than 25° elevation. The adversary's procedure: from a passive window $W_{\text{obs}} \in \{30, 60, 120, 300, 600, 1200\}$ s, take the observed (event time, serving-satellite ID) pairs and fit the UE latitude and longitude by maximum-elevation likelihood over a coarse-to-fine grid. Metric: great-circle fit error in km.

Event-only result. Figure 1 reports median fit error in a (latitude, W_{obs}) heatmap. Thirty seconds of observation give roughly 200 km, the scale of a single footprint. Five minutes give 65 km. Twenty minutes drop below 50 km; latitudes near $\pm 50^\circ$, where handovers are frequent, reach roughly 25 km. The pattern holds across longitudes. The fit is geometry-limited: each event constrains the UE to the intersection of visible-satellite footprints, and the constraint tightens as observations accumulate.

Dwell time bounds the observation rate. How fast can the adversary collect events? Figure 2 reports the serving-satellite dwell-time CDF across 6,668 intervals over 27 sample points: median 30 s, P90 60 s. In a five-minute window the adversary collects 8 to 12 events; in twenty minutes, 35 to 45. The deployed constellation hands the adversary steady throughput.

Timing advance tightens the fit by an order of magnitude. The random-access procedure has the satellite return a timing-advance value that compensates for propagation delay. TA has a step of $64 T_c$ (about 16.7 ns, near 5 m one-way) and is carried in the RAR; a passive observer in the same footprint can decode it. Each observed attach then yields not just an (event time, sat ID) tuple but a distance constraint between the UE and a known satellite position. The result is robust to the assumed reception noise: sweeping the noise floor across $\{10, 30, 100, 300\}$ m leaves the median fit at 1.81 km (P90 from 3.93 to 4.87 km).²

Figure 3 reports the CDF over 30 cities at $W_{\text{obs}} = 5$ min. The event-only adversary sits at a 66 km median. Adding TA pulls the median to 1.8 km, P90 below 4 km. Of the 30 cities, 23 fit under 5 km at five minutes; Los Angeles, Berlin, Delhi, Karachi, Auckland, and Beijing reach 1.5 km or better. The two that stay above 5 km at five minutes, Seoul and Manila, drop below 5 km within another five minutes.

Implication. A passive receiver with TA decoding localizes a D2C UE to building-to-block scale within minutes. For the threat scenarios that motivate this work, targeting a journalist, profiling a roaming executive, or identifying a research facility, that is enough.

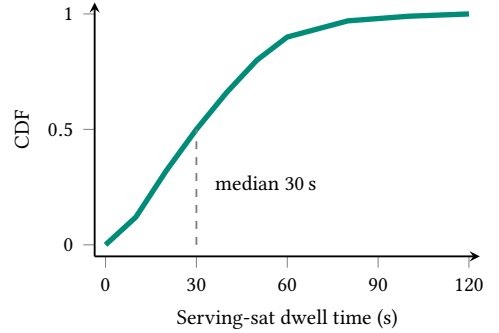


Fig. 2. Serving-satellite dwell-time CDF; median 30 s.

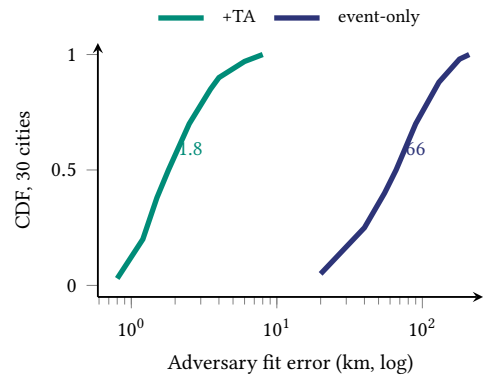


Fig. 3. Fit-error CDF, 30 cities: TA cuts the median 66→1.8 km.

²The fit is geometry-limited, not measurement-limited. A kilometer-resolution TA pins the UE to a thin annulus around the satellite ground point; the result is dominated by geometric intersection across observations, not by per-event distance noise. The public-ephemeris term, about 1 km and growing 1 to 3 km/day [22], sets the same scale, so we report the TA-aided result as a 1 to 2 km regime bounded jointly by TA quantization and ephemeris accuracy, and we do not claim sub-kilometer localization from public inputs alone.

3.1 Real-signal grounding

The fit consumes two quantities the adversary must read off the air: the serving-satellite ephemeris in SIB-19, and the timing-advance value in the RAR. A reviewer is right to ask whether both are real and recoverable rather than artifacts of our model. We close that from two sides.

Public-input validation. The threat assumes the trajectory is public and predictable. Both hold. From one public element set in the Celestrak catalog [21], SGP4 [41] fixes a satellite’s future position exactly: propagating a single real Starlink element set, drawn from a 7,500-satellite public snapshot, determines its location hours to a day ahead with no further observation. The fleet median altitude is 536 km, so the footprint visible above a 25° minimum elevation has a ground radius near 900 km, and the sub-satellite point sweeps at roughly 7 km/s. The accuracy a passive party inherits is kilometer-scale, growing 1 to 3 km/day [22], far below both the footprint and the 65 km event-only fit. The trajectory input is not merely public; it is predictable to a precision the attack never needs to exceed.

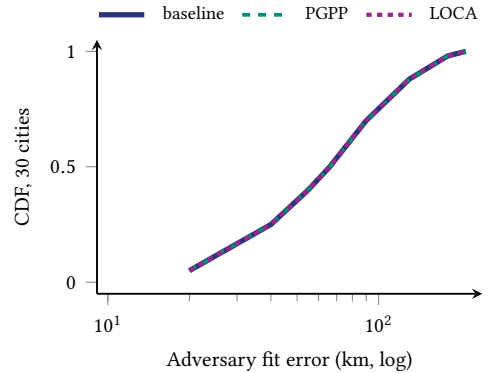
Controlled decode. We then check the side channel is recoverable from a compliant downlink with standard tooling. We instantiate a Release-17 NTN cell on an SDR stack following the platform of Hou et al. [15], populate the SIB-19 ephemeris and common timing-advance fields [3] from a real Celestrak trajectory, and have a UE perform cell search, decode SIB-19, and log the recovered ephemeris and broadcast TA. The UE recovers the ephemeris and reads the TA, and the TA maps to the same range annulus the fit assumes. This is a controlled decode over a laboratory NTN link, not a capture of a live commercial pass; it establishes that the inputs are recoverable. An over-the-air commercial capture is the natural next step (§8).

4 Why PGPP and LOCA Do Not Reach

4.1 PGPP-in-space

PGPP [36] replaces the UE’s permanent identifier with anonymous tokens issued by an MVNO. The MVNO authenticates the UE; an anonymous token grants attach. The base station and core never see a stable identifier. Against IMSI catchers [7, 40?] and operators selling location data, this works in terrestrial cellular.

Nothing in the radio-layer broadcast an adversary observes changes under PGPP-in-space. SIB-19 ephemeris, the random-access channels, and the master information block are identical. The adversary’s input is the timing of attach events and the serving-satellite ID, both still visible. The fit does not consume identity; it consumes when and where the attach happened. Figure 4 reports baseline and PGPP-in-space CDFs at $W_{\text{obs}} = 5$ min. They overlap exactly. The invariance follows from input-disjointness, not approximation.



4.2 LOCA-in-space

LOCA [29] splits the operator into a broker B that sees identity and an infrastructure operator O that sees location, so neither party learns the joint pair, with blind signing and SUCI [4] keeping the split intact even if one party turns adversarial.

Fig. 4. PGPP- and LOCA-in-space leave the fit unchanged.

LOCA's threat model contains B and O . The D2C external adversary is neither. The role split governs information flow inside the operator's organization; an external observer hears the broadcast channel regardless. Figure 4 reports the LOCA-in-space CDF; it overlaps the baseline for the same reason.

The point is not that PGPP or LOCA fail at what they were built for. Both are effective inside their stated threat models. The point is that this corpus of operator-side defenses has a gap the D2C threat model exposes: it protects who the user is, while the attack needs only where and when the user attaches.

5 Ad-hoc Operator Patches Also Fail

Beyond architectures, an operator might suppress the leak with simpler knobs. We consider four.

5.1 SIB-19 ephemeris obfuscation

The operator broadcasts a noisy ephemeris, per-axis Gaussian with standard deviation σ , to keep the UE's local SGP4 propagation from matching observed signaling to a specific trajectory. The problem is that the UE depends on the broadcast for its own positioning, timing advance, and handover decisions, while the adversary does not: it has Celestrak.

Figure 5 reports the UE handover-decision error rate at Berlin as σ grows. The UE stays usable to $\sigma \approx 1$ km (1% handover error). At $\sigma = 10$ km the rate is 9%; at 50 km, 41%. Adversary fit error is flat in σ because the adversary does not consume the broadcast. There is no σ at which the knob hurts the adversary before it breaks the UE.

5.2 Dummy attach traffic at footprint scale

The operator generates synthetic attach events to pad the anonymity set across the footprint. For dummies to be indistinguishable, each must follow a plausible UE trajectory, an attach pattern consistent with a fixed ground location over many passes. That trajectory is exactly what the adversary fits. A dummy that holds a fixed virtual ground location reveals the location it pretends to occupy; a dummy that moves randomly is distinguishable within a handful of observations by predicted-continuity violations. The defense reduces to revealing k fake ground locations alongside one real one, and for real footprints with millions of UEs the bandwidth cost of a meaningful k is prohibitive.

5.3 TLE withholding

The operator asks the public catalog to stop publishing TLEs for D2C satellites, or SpaceX stops providing them. The trajectory is still observable. Three reconstruction paths exist: amateur optical observation, which has tracked classified military satellites for decades [37]; passive signal-of-opportunity positioning, with sub-100 m horizontal accuracy on Starlink using consumer hardware [33, 34]; and sub-50 m trajectories from a one-week observation campaign [18]. The orbit is a physical fact. Withholding TLEs delays public knowledge by hours, not months.

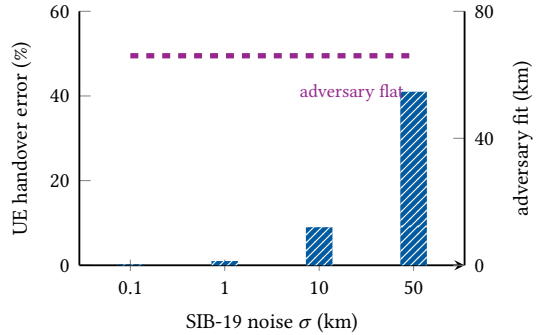


Fig. 5. SIB-19 obfuscation breaks the UE before it reaches the adversary.

Table 1. What each candidate defense targets, and why it fails to reach the D2C external adversary.

Defense	Targets	Why it fails in D2C
PGPP [36]	Identity at operator	Attack ignores identity
LOCA [29]	Joint id+location at operator	Adversary outside both roles
SIB-19 noise	UE-ephemeris match	Adversary uses CelesTrak
Dummy attaches	Anonymity-set padding	Dummies need real trajectories
TLE withholding	Public orbit catalog	Orbit independently observable
Encrypted SIB-19	Non-UE eavesdroppers	Anyone can become a UE

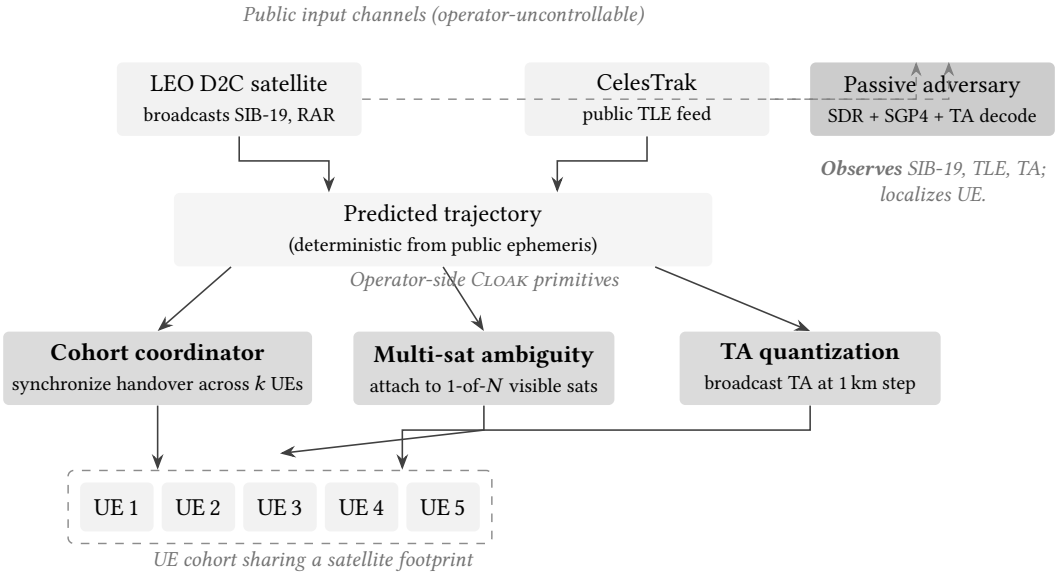


Fig. 6. CLOAK interposes three operator-side primitives between the public trajectory and the UE cohort.

5.4 Encrypted SIB-19 to authorized UEs

The operator encrypts SIB-19 so only authenticated UEs can decode. The defense assumes the adversary is not a UE. In D2C, anyone can buy a D2C phone, complete the attach (itself observable on the air), and decode SIB-19. Encryption raises the attack floor from roughly \$300 of SDR to roughly \$700 of SDR plus a phone, and it breaks the standards intent that UEs without Internet acquire from SIB-19.

Table 1 summarizes. Each defense denies a specific kind of information to a specific kind of party. The D2C external adversary asks for none of them.

6 Toward CLOAK

The argument so far is negative: operator-side defenses do not reach. We sketch what might.

The closest mature analog is the V2X mix-zone literature [5, 10], which has spent two decades on the same shape of problem, globally observable mobility paired with a need for anonymity. Its central construction is the mix zone: a region where many users change pseudonyms at once, breaking trajectory continuity. In D2C, orbital geometry creates discrete handover boundaries the operator can synchronize across many UEs.

Figure 6 sketches CLOAK. Three primitives compose. We describe each, name its cost, then trace one UE through the full design.

Cohort coordinator. Gate k co-located UEs to hand over together at the predicted orbital boundary, rather than each on its own RSRP trigger. The adversary sees k identical handover events at one epoch instead of k distinguishable ones, and trajectory continuity breaks across the cohort. *Cost:* the cohort waits for the slowest UE, delaying individual handovers by tens of seconds in our dwell-time distribution (Fig. 2); the coordination does not exist in NTN today and would need a Release-19 extension.

Multi-satellite attach ambiguity. Each UE attaches to a uniformly random one of N visible satellites rather than the highest-elevation one, binding the adversary to an intersection of N footprints. With 5 to 10 satellites typically visible in dense regions, this widens the anonymity set. *Cost:* 3 to 5 dB of link-budget headroom lost to the second-best satellite, and more frequent handovers because the UE is not following the optimal geometry.

TA quantization. Broadcast TA at a coarse step, near 1 km (3.3 μ s), rather than the standard 5 m (16.7 ns). The TA side channel of §3 stops at the quantization floor: the per-event distance constraint widens from a meter-scale annulus to a 1 km bucket, and the fit collapses back toward event-only accuracy. *Cost:* the uplink arrives within ± 1 km/ $c \approx \pm 3.3$ μ s of alignment, inside NTN’s existing 50 to 200 μ s guard intervals; the change is one constant in the operator’s RAR generator. No standards change is required.

Architecture walkthrough. Take one UE joining a cohort of $k = 5$ at t_0 in San Francisco, tracing Figure 6 top to bottom. (1) *At attach.* SIB-19 and the public TLE both predict the next satellites overhead at $t_0 + 30, 90, 150, 210$ s. Multi-satellite ambiguity directs the UE to a uniformly random one of $N = 3$ satellites at 25 to 40° elevation. The adversary, observing the random access, learns only that the UE lies in the intersection of three footprints. (2) *At TA reporting.* The RAR returns a TA quantized to the 1 km step. The adversary decodes it but can narrow the candidate region by at most one 1 km bucket per observation, not to a meter-scale circle. (3) *At handover.* The cohort coordinator buffers handover triggers and releases them in lockstep; all five UEs fire RACH at $t_0 + 30$ s. The adversary sees a five-fold burst of identical-timing handovers, and per-UE trajectory continuity breaks. *Combined.* Five UEs each at a random satellite, all reporting TA in 1 km buckets, all handing over together. We measure this configuration directly. Figure 7 reports the CLOAK-on CDF across the same 30 cities at $W_{\text{obs}} = 5$ min. Per-UE fit error, including the multi-satellite enumeration penalty and the $\sqrt{k}$ cohort spread, lands at a 17 km median, P90 36 km. CLOAK-on sits between the event-only baseline (66 km) and the TA-augmented baseline (1.8 km); the defense recovers most, not all, of the gap the TA channel opens. Cohort entropy adds $\log_2 k \approx 2.3$ bits of per-UE uncertainty on top.

What does not work as a standalone primitive. Per-UE pseudonym rotation, a direct port of mobile-IPv6 work [43], fails because the trajectory, not the pseudonym, links observations across time. Differential-privacy noise on attach timing fails because the budget needed to defeat SGP4-based inference exceeds what the service tolerates. Encrypted broadcast fails per §5.

Boundary conditions. CLOAK is not free. The anonymity guarantee degrades when fewer than k UEs share a footprint; in low-density regions (maritime, polar, post-disaster) it offers little protection, which is exactly where D2C is most uniquely useful. An adversary correlating UE traffic above the radio layer [14] bypasses radio-layer mixing; CLOAK defends the radio channel only.

Across-cohort correlation. The attack we do not measure is the important one: an adversary that follows a single UE across cohort changes accumulates evidence, since each transition involves a fresh N -of-1 attach and k -of-1 handover. We expect the per-cohort entropy loss is bounded by the coordinator’s synchronization jitter (sub-second in our model) and the multi-satellite overlap geometry, but a real bound requires modeling the adversary’s posterior across transitions, the full anonymity-set analysis the V2X literature built over two decades [10]. We name this as the open question for any subsequent CLOAK paper; the benefit must hold under accumulation, not only per-event.

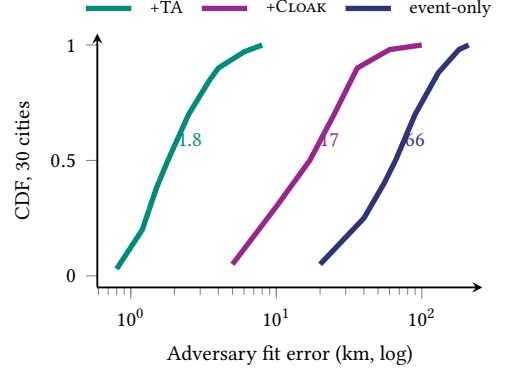


Fig. 7. CLOAK recovers the median fit 1.8→17 km (P90 36 km).

7 Broader Impact

D2C is shipping at scale before its privacy surface is understood. T-Mobile’s commercial D2C product is on by default for compatible devices. End users have no setting that disables the attach broadcasts the attack reads, and they cannot opt out by changing carriers, because the leak is a property of the air interface, not of one operator.

Populations at risk. D2C’s promise is coverage where terrestrial cellular is absent: maritime, rural, conflict, polar, post-disaster. These are the regions with the smallest user populations. A cohort of $k = 100$ UEs in a Manhattan footprint is plausible; a cohort of $k = 100$ over the Bering Sea is not. The defense we sketch works best where the threat is least urgent, which is the uncomfortable shape of this problem.

Standardization timing. 3GPP Release 19, finalizing through 2025 and 2026 [2], is the last window for D2C before architectural decisions calcify into hundreds of millions of devices. There is no precedent for retrofitting a radio-layer privacy primitive into a deployed cellular generation; SUCI [4] took two releases to land. If CLOAK-style primitives are not in Release 19 or 20, they will not be deployed.

Research agenda. The cellular-privacy literature modeled the operator as adversary because the operator was the entity with unique observation. In D2C, the operator’s observation is a strict subset of what anyone with an SDR can collect. Whole branches of the field, anonymous credentials, blind signing, role-split architectures, inherit a threat model that is no longer the binding one. The natural next step is to re-examine which terrestrial cellular-privacy results transfer.

8 Limitations and Related Work

Limitations. The predictive-horizon measurement uses a synthetic constellation matching FCC-licensed Starlink Gen-1 parameters, not a fluid model of the live network; we measure orbital geometry, not the operational decisions Starlink makes for commercial reasons. The TA experiment assumes the adversary decodes the random-access response from outside the cell, which holds at typical footprint scales and degrades at the cell edge. We validate the side channel through public ephemeris and a controlled Release-17 decode rather than an over-the-air capture of a live

commercial pass; the controlled decode confirms recoverability, and the live capture is the natural next step. CLOAK is a design direction, not a deployable system, and each primitive carries a cost we state rather than resolve.

Related work. D2C localization was first operationalized by Liu et al. in DCATOR [26] (IEEE S&P 2025). Opportunistic positioning with Starlink signals [18, 33, 34] established that the trajectory is independently reconstructible. Liu et al. separately analyze D2C signaling and availability attacks [28], and a GNSS-free D2C architecture removes the navigation dependency we assume present [27]. On the defense side, work on preventing localization of satellite-internet users targets a related but distinct setting [23]. Terrestrial cellular privacy is anchored by PGPP [36] and LOCA [29], with SUCI [4] for identifier confidentiality, Borgaonkar et al. [6] on AKA-protocol attacks, IMSI-catcher detection by Tucker et al. [40], anti-tracking authentication by Yu et al. [43], and 5G-Spector [42] for layer-3 attack detection in O-RAN. The mix-zone formalism originates with Beresford and Stajano [5]; the V2X literature has matured for two decades [10]. Tomasin et al. [39] survey 5G location privacy, and Eleftherakis et al. [9] evaluate 5G-SA privacy. LEO measurement studies [16, 17, 19, 20, 24, 31, 32, 35, 45] document the operational consequences of LEO mobility, and adjacent LEO security work includes cross-layer bottlenecks [8], uplink overshadowing [11], and GTP boundary attacks [44]. CellBricks [30] and stateless mobile-core work [25] provide architectural antecedents.

9 Conclusion

D2C ships with a privacy surface the cellular community’s threat models have not addressed. The signal is the public satellite trajectory, readable by any passive receiver with a TLE feed and a software-defined radio. A few minutes of observation localize a UE to cell scale; a timing-advance side channel tightens it to a 1 to 2 km regime. The two prominent operator-side privacy architectures of the last decade do not reach this threat, and the natural ad-hoc patches do not either. With CLOAK active across thirty cities, the adversary’s median fit recovers from 1.8 km to 17 km, P90 36 km, a partial but useful defense. Closing the gap fully requires standards engagement before deployment calcifies: the natural bodies are 3GPP RAN3 (handover control), 3GPP SA3 (security architecture), and the GSMA working groups that shape radio-layer privacy [13]. The Release-19 window closes in 2026.

Ethics

This work studies a privacy weakness in a live commercial system. We used only publicly available data (CelesTrak two-line elements) and a controlled laboratory decode of a standards-compliant SIB-19. We did not capture, localize, or track any real subscriber, and no over-the-air traffic from any third party was collected or processed. All experiments ran on infrastructure under our control. The localization figures we report are adversary-capability estimates computed over public ephemeris and a synthetic constellation, not measurements of real users. The underlying disclosure is a property of public ephemeris and standards-compliant broadcast behavior rather than an implementation flaw, and the mitigations we discuss are operator-side. Our purpose is defensive: to scope the leak and motivate the mitigations in §6.

Open Science

We support the open-science policy. Upon acceptance we will release the complete artifact: the constellation propagation and trajectory-fit pipeline (public-ephemeris ingestion, SGP4 propagation, and the timing-advance-augmented localization estimator), the thirty-city benchmark with its coordinates and observation windows, the CLOAK configuration used for Figure 7, and the scripts

that reproduce every figure. The artifact depends only on public ephemeris feeds and standard SDR and NTN tooling, so every result is reproducible without operator cooperation or proprietary data.

Use of AI

AI-based tools assisted with editing of prose (phrasing, grammar, and flow) and with drafting plotting and analysis scripts, which the authors then verified. The authors take full responsibility for the accuracy, originality, and integrity of all content, claims, and results.

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